

WEEK 1

GITHUB WORKFLOW & BASIC STATS

APPLIED STATISTICAL ANALYSIS/QUANTITATIVE METHODS I

LECTURER: JEFFREY ZIEGLER, PHD

TEACHING FELLOWS:

ELENA KARAGIANNI

SHEKHAR KEDIA

ASDS - TRINITY COLLEGE DUBLIN

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ROAD MAP FOR TODAY

- Resolve any outstanding R/Python set-up issues
- Get familiar with GitHub
- Mock problem set upload
- Review basic stats concepts

VERSION CONTROL AND GIT

- Version control systems (VCSs) allow automatic tracking of changes in files and collaboration
- Git is one of several major version control systems (VCSs, see also Mercurial, Subversion)
- GitHub is an online hosting platform for projects that use Git for version control

LET'S CLEAR UP SOME THINGS

Git and GitHub sound similar, but are different things:

- Git is the tool that makes version history on your computer
- GitHub is like Google Drive; it lets you sync and share those histories online

- Hosting platform for projects that rely on Git for version control
- Bought by Microsoft in 2018
- Provides extensive tools for collaborative development and search functionality
- Helpful for troubleshooting more narrow problems
 - ▶ Check GitHub issues for the package or library with which you have a problem
- GitHub is far from the only platform for hosting Git projects

SOME USEFUL GIT COMMANDS TO REMEMBER

For a more detailed cheatsheet:

[Git Cheatsheet](#)

USING GITHUB AND GITHUB DESKTOP (EASIER)

- Set up a GitHub account at: <https://github.com/join>
- You need to fork the parent repository
 - ▶ This makes your own copy of the folder
- Go to the main folder StatsI2026 and click *Fork*
- You will now have your own repository that you control, linked to the parent repository
- Install GitHub Desktop: <https://desktop.github.com/>

LINK ONLINE REPOSITORY TO LOCAL REPOSITORY

- With the desktop version downloaded, let's link your online repository with your local repository
- Click the green *Code* button
- Select the option to open the repository in GitHub Desktop

LINK ONLINE REPOSITORY TO LOCAL REPOSITORY

- You will be transferred over to GitHub Desktop
- Make sure that your folder is placed in:
 - 1 Documents / GitHub / StatsI2026
- If asked, do not contribute to the parent directory
- You now have a completely up-to-date version of the files from parent directory
- You also have a synchronised folder between your local files and your online repository

POSTING HOMEWORK TO GITHUB

- Suppose you finish your homework and want to post your answers to GitHub
- Assuming that you are editing your files in your local GitHub folder, GitHub Desktop will automatically recognise changes that you have made
- Write a summary to remind yourself what changes you have made
- For example:

```
1 Post PSo1
```

- Then click *Commit to main*

PUSHING CHANGES

- Committing puts your changes in Git purgatory
- You need to finish by selecting *Push to origin*
- This synchronises your local files with your online files

FETCHING NEW FILES

- You will need to get updated homework from the parent repository
- You will see that there are files that you do not have
- You will therefore need to synchronise your repositories
- Use GitHub Desktop to synchronise your online repository with your local files

FETCHING NEW FILES FROM ORIGIN

- To synchronise, you need to fetch from origin
- You will see that there are files that are different

PULLING TO ORIGIN

- Once you fetch from origin, GitHub Desktop will identify files that are different
- In this instance, there are four different files
- Click *Pull origin*
- This adds the missing files from your online repository to your local folder
- Check your Documents folder to confirm that the files have been added

LET'S TRY TO ADD YOUR OWN MOCK PS

REVIEW: BASIC STATS

Ex: Calculating mean, variance, and SD

y_i	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$
0		
4		
4		
5		
7		
10		

DOING IT IN R: FIND THE SUM

```
1 # create vector y
```

DOING IT IN R: FIND THE SUM

```
1 # create vector y  
2 y <- c(0, 4, 4, 5, 7, 10)
```

DOING IT IN R: FIND THE SUM

```
1 # create vector y
2 y <- c(0, 4, 4, 5, 7, 10)
3 # (1) find sum of y
```

DOING IT IN R: FIND THE SUM

```
1 # create vector y
2 y <- c(0, 4, 4, 5, 7, 10)
3 # (1) find sum of y
4 sum(y)
```

$$\sum y_i = 30$$

DOING IT IN R: FIND THE MEAN

The old fashioned way:

$$\frac{\sum y_i}{n} \quad (1)$$

DOING IT IN R: FIND THE MEAN

```
1 # (2) find mean of y  
2 sum(y)/length(y)
```

DOING IT IN R: FIND THE MEAN

```
1 # (2) find mean of y
2 sum(y)/length(y)
3 # same thing, but faster
4 mean(y)
```

$$\bar{y} = 5$$

DOING IT IN R: FIND THE SUM OF ERRORS

Let's try to construct this ourselves:

$$\sum (y_i - \bar{y})$$

DOING IT IN R: FIND THE SUM OF ERRORS

```
1 # (3) find sum of demeaned values
2 # subtract mean from each y_i
3 # first, I'll create a vector to fill
4 # there is nothing in there when I create an object
5 # like x <- NULL
6 demeanedSum <- NULL
7 # remember that [i] is indexing each element of y
8 # I'm basically saying, for each element in all of the
  elements in y
9 # take that y_i and subtract the mean of y
10 # and store that value in my vector of demeaned values for
  each y_i
11 for(i in 1:length(y)){
12   demeanedSum[i] <- y[i] - mean(y)
13 }
```

[1] -5 -1 -1 0 2 5

FILLING IN THE TABLE WITH OUR DEMEANED VALUES

y_i	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$
0	-5	
4	-1	
4	-1	
5	0	
7	2	
10	5	

DOING IT IN R: FIND THE SUM OF ERRORS

```
1 # the simple way (R figures it out)
2 demeanedSumSimple <- y - mean(y)
3 # now we take the sum of all those demeaned values
4 sum(demeanedSumSimple)
```

$$\sum (y_i - \bar{y}) = 0$$

FINISHING SUM OF SQUARED ERRORS IN R

```
1 # (4) sum of squared error
2 squaredError <- demeanedSum^2
3 sum(squaredError)
```

$$\sum (y_i - \bar{y})^2 = 56$$

FINISHING OUR TABLE WITH THE SQUARED ERRORS

y_i	$(y_i - \bar{y})$	$(y_i - \bar{y})^2$
0	-5	25
4	-1	1
4	-1	1
5	0	0
7	2	4
10	5	25

ANSWER: CALCULATING VARIANCE

$$s^2 = \frac{\sum (y_i - \bar{y})^2}{n - 1}$$

ANSWER: CALCULATING VARIANCE

$$S^2 = \frac{\sum (y_i - \bar{y})^2}{n - 1}$$

$$S^2 = \frac{56}{5} = 11.2$$

ANSWER: CALCULATING STANDARD DEVIATION

$$S^2 = \frac{\sum (y_i - \bar{y})^2}{n - 1}$$

$$S^2 = \frac{56}{5} = 11.2$$

$$S = \sqrt{11.2} = 3.3$$

CLASS BUSINESS

- Read required (and suggested) online materials
- Fork GitHub repository (if it didn't work during tutorial, please ask Jeff!)
- Problem set # 1 is up on GitHub